

Mini Project 3: Recurrence-Free Breast Cancer Survival

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Mini Project — Overview

In this mini-project, you will analyze recurrence-free survival data from the German Breast Cancer Study Group 2 **GBSG2 study** using Kaplan–Meier curves and Cox proportional hazards models.

Your goal is to:

- Explore the data
- Build and compare Cox models
- Investigate an interaction
- Interpret model results
- Justify your final model

Submit a single `.qmd` file and rendered PDF including:

- Code
- Output
- Written answers (concise but clear)

Data Description

```
data("GBSG2", package = "TH.data")
summary(GBSG2)
```

horTh	age	menostat	tsize	tgrade
no :440	Min. :21.00	Pre :290	Min. : 3.00	I : 81
yes:246	1st Qu.:46.00	Post:396	1st Qu.: 20.00	II :444
	Median :53.00		Median : 25.00	III:161
	Mean :53.05		Mean : 29.33	
	3rd Qu.:61.00		3rd Qu.: 35.00	
	Max. :80.00		Max. :120.00	

pnodes	progrec	estrec	time
Min. : 1.00	Min. : 0.0	Min. : 0.00	Min. : 8.0
1st Qu.: 1.00	1st Qu.: 7.0	1st Qu.: 8.00	1st Qu.: 567.8
Median : 3.00	Median : 32.5	Median : 36.00	Median :1084.0
Mean : 5.01	Mean : 110.0	Mean : 96.25	Mean :1124.5
3rd Qu.: 7.00	3rd Qu.: 131.8	3rd Qu.: 114.00	3rd Qu.:1684.8
Max. :51.00	Max. :2380.0	Max. :1144.00	Max. :2659.0

cens
Min. :0.0000
1st Qu.:0.0000
Median :0.0000
Mean :0.4359
3rd Qu.:1.0000
Max. :1.0000

```
GBSG2 = GBSG2 %>%
  mutate(tgrade = factor(tgrade))
```

Key variables:

- **time**: recurrence-free survival time (days)
- **cens**: event indicator (1 = recurrence, 0 = censored)
- **horTh**: hormonal therapy (yes/no)
- **age**: age in years
- **menostat**: menopausal status
- **tsize**: tumor size
- **tgrade**: tumor grade (1, 2, or 3)
- **pnodes**: number of positive nodes
- **progrec, estrec**: hormone receptor levels

Part 1 — Exploratory Analysis

1. Overall Kaplan–Meier Curve

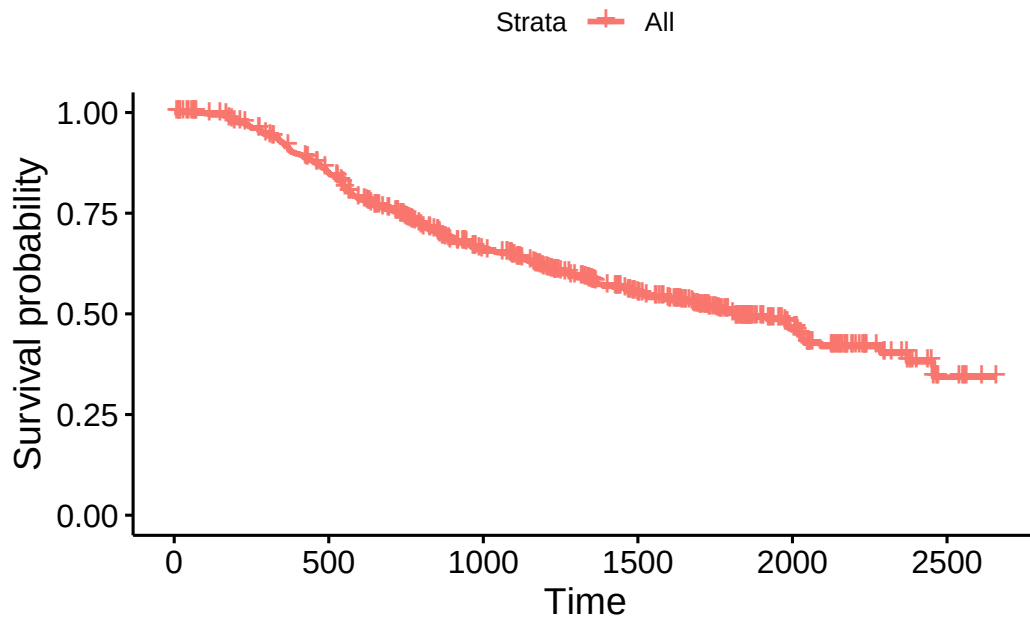
1. Plot Kaplan–Meier curves.

```
library(survival)
fit <- survfit(Surv(time, cens) ~ 1, data = GBSG2)

# -----
# Plot KM curves
# -----

out <- ggsurvplot(
  fit,
  data = GBSG2,
  risk.table = TRUE,
  conf.int = FALSE,
  xlim = c(0, 2660),
  linetype = "strata"
)

out$plot
```



2. The median survival time is the time at which the survival probability is 50%. Based on the KM-Curve, what is the approximate median recurrence-free survival time in the data?

The median occurs at around 1750 days.

2. Kaplan–Meier Curve by Group

1. Plot Kaplan–Meier curves for:

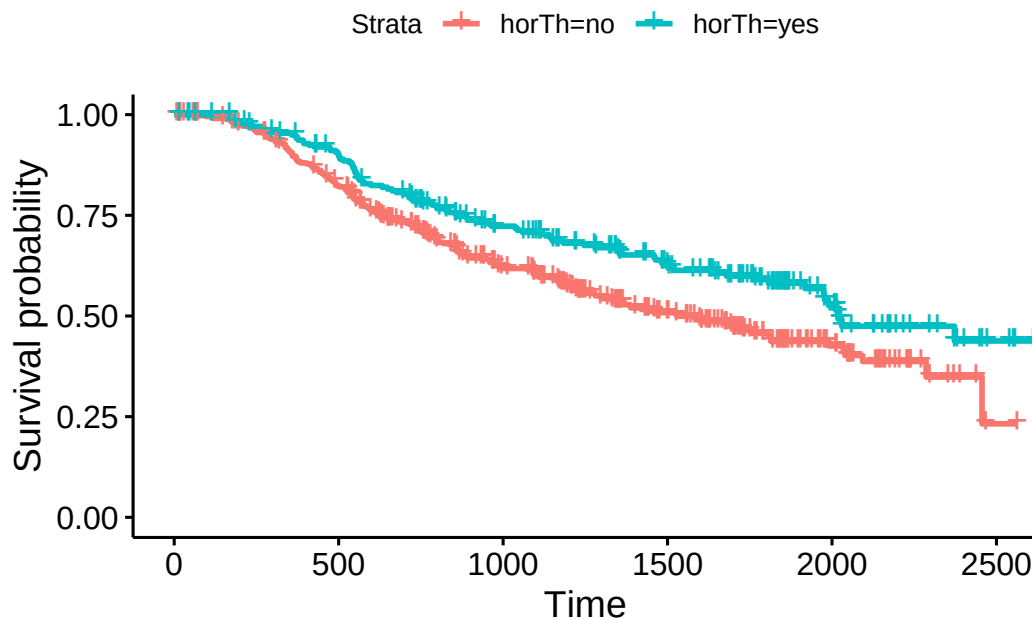
- horTh

#Update the generic code below:

```
fit = survfit(Surv(time, cens) ~ horTh, data = GBSG2)

out = suppressWarnings(ggsurvplot(fit,
  data = GBSG2,
  conf.int = FALSE))

out$plot
```



2. Briefly describe differences in survival across groups.

There isn't a super pronounced difference in the survival between the groups at first, but as time goes on it becomes clear that the patients that received hormone therapy consistently went longer before breast cancer recurrence.

3. Covariate Exploration

Explore relationships between predictors and survival.

1. Compare distributions of two of the following variables between those who survived and those who didn't (`cens = 0` vs. `cens = 1`):

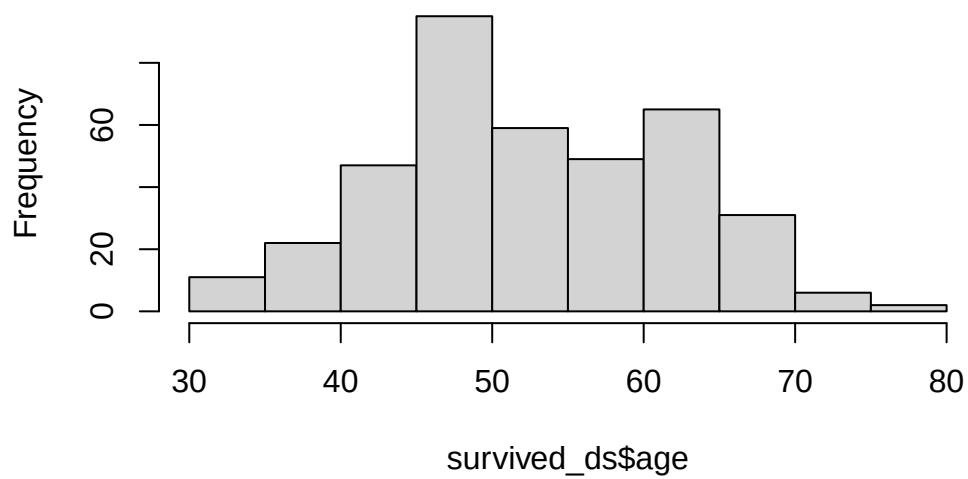
- `age`
- `tsize`
- `pnodes`
- `estrec`
- `progre`

```
survived_ds <- GBSG2 |>
  filter(cens == 0)

returned_ds <- GBSG2 |>
  filter(cens == 1)

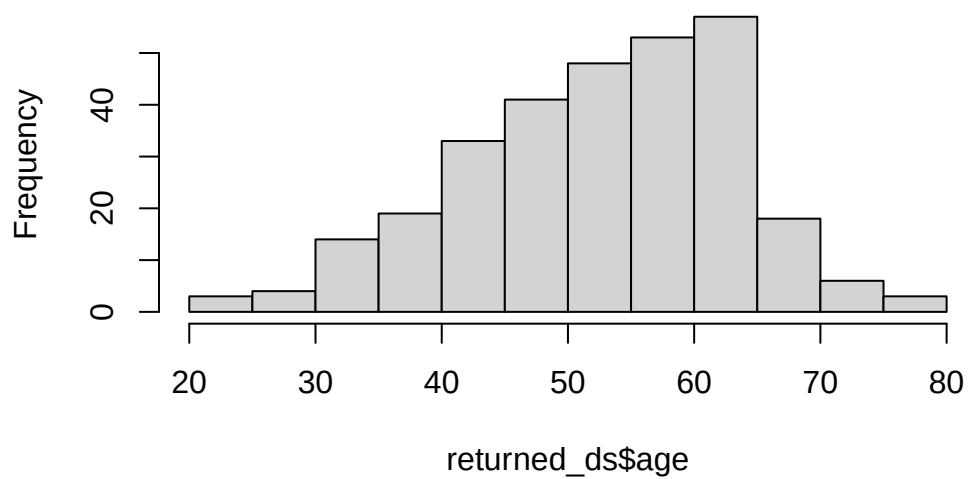
hist(survived_ds$age)
```

Histogram of survived_ds\$age



```
hist(returned_ds$age)
```

Histogram of returned_ds\$age



```
hist(survived_ds$tsize)
```



```
hist(returned_ds$tsize)
```



2. What patterns do you notice?

I notice that the distribution of ages tends to lead older among those for whom the breast cancer returned, while the distribution of the sizes of the tumor doesn't seem to be very different between the group that survived and the group where the breast cancer returned.

Answer Here

3. Reflect on how censoring might impact these simple distributional comparisons. What might we miss by only looking at these summaries?

- hint: Think about whether a 'survivor' ($\text{cens} = 0$) has necessarily lived longer than someone with a recurrence ($\text{cens} = 1$).

There's a lot that we might miss by assuming that censored ($\text{cens} = 0$) indicates survival, because there are many reasons why someone might drop out of the study besides the breast cancer going away. For instance, they might have switched to a different doctor, moved, simply got too busy to respond, or they might've gone looking for another treatment. Any one of these could mess with our interpretation if we're treating $\text{cens} = 0$ as a survival.

Part 2 — Cox Model Building

4. Baseline Cox Model

Fit a Cox model using:

- age
- pnodes
- horTh

```
fit1 <- coxph(Surv(time, cens) ~ age + pnodes + horTh, data = GBSG2)
summary(fit1)
```

Call:

```
coxph(formula = Surv(time, cens) ~ age + pnodes + horTh, data = GBSG2)
```

```
n= 686, number of events= 299
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
age	-0.0008626	0.9991377	0.0059934	-0.144	0.88556

```

pnodes      0.0577092  1.0594068  0.0066618  8.663  < 2e-16 ***
horThyes    -0.3529801  0.7025911  0.1283208 -2.751  0.00595 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

      exp(coef) exp(-coef) lower .95 upper .95
age      0.9991      1.0009      0.9875      1.0109
pnodes    1.0594      0.9439      1.0457      1.0733
horThyes   0.7026      1.4233      0.5464      0.9035

```

```

Concordance= 0.654 (se = 0.016 )
Likelihood ratio test= 58.51 on 3 df,  p=1e-12
Wald test               = 85.51 on 3 df,  p=<2e-16
Score (logrank) test = 88.15 on 3 df,  p=<2e-16

```

1. Report exponentiated coefficients (hazard ratios).

```

# You can use the following if you want
tidy(fit1, exponentiate = TRUE)

```

```

# A tibble: 3 x 5
  term      estimate std.error statistic  p.value
<chr>      <dbl>      <dbl>      <dbl>    <dbl>
1 age        0.999      0.00599     -0.144 8.86e- 1
2 pnodes     1.06       0.00666      8.66 4.61e-18
3 horThyes   0.703      0.128      -2.75 5.95e- 3

```

2. Interpret two exponentiated coefficients in context – one for a continuous variable and one for a categorical variable.

The coefficient for pnodes tells us that every extra positive node that someone has means that their risk at all points in time goes up by 5.94%, holding all else constant.

The coefficient for horThyes tells us that those that get hormone therapy have 70.26% of the risk at all points in time that people that don't get hormone therapy have, holding all else constant.

5. Extended Model

Fit a more complex model by adding additional predictors (e.g., `tgrade`, `menostat`, `tsize`, or receptor variables).

```
fit2 <- coxph(Surv(time, cens) ~ age + pnodes + horTh + tgrade + menostat + tsize, data = GBSG2)
summary(fit2)
```

Call:

```
coxph(formula = Surv(time, cens) ~ age + pnodes + horTh + tgrade +
      menostat + tsize, data = GBSG2)
```

n= 686, number of events= 299

	coef	exp(coef)	se(coef)	z	Pr(> z)
age	-0.012375	0.987701	0.009019	-1.372	0.170024
pnodes	0.050282	1.051568	0.007414	6.782	1.18e-11 ***
horThyes	-0.358227	0.698915	0.128775	-2.782	0.005406 **
tgrade.L	0.707490	2.028892	0.185691	3.810	0.000139 ***
tgrade.Q	-0.213694	0.807595	0.121791	-1.755	0.079329 .
menostatPost	0.332118	1.393918	0.180867	1.836	0.066320 .
tsize	0.007325	1.007352	0.003887	1.884	0.059524 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age	0.9877	1.0125	0.9704	1.0053
pnodes	1.0516	0.9510	1.0364	1.0670
horThyes	0.6989	1.4308	0.5430	0.8996
tgrade.L	2.0289	0.4929	1.4099	2.9196
tgrade.Q	0.8076	1.2382	0.6361	1.0253
menostatPost	1.3939	0.7174	0.9779	1.9870
tsize	1.0074	0.9927	0.9997	1.0151

Concordance= 0.674 (se = 0.015)

Likelihood ratio test= 83.44 on 7 df, p=3e-15

Wald test = 101.2 on 7 df, p=<2e-16

Score (logrank) test = 107.2 on 7 df, p=<2e-16

Compare this model to the baseline model using:

- AIC or likelihood ratio test

```
AIC(fit1, fit2)
```

```
      df      AIC
fit1  3 3523.700
fit2  7 3506.771
```

- Which model is better? Why?

It looks like the second model has a slightly lower AIC, but it's not by a lot so it might be worth taking the first one if you want a simpler model.

6. Interaction

Add **one interaction term** of your choice.

Examples: - horTh * menostat - age * pnodes

```
fit3 <- coxph(Surv(time, cens) ~ age + pnodes + horTh + tgrade + menostat + tsize + age * horTh)
summary(fit3)
```

Call:

```
coxph(formula = Surv(time, cens) ~ age + pnodes + horTh + tgrade +
      menostat + tsize + age * horTh, data = GBSG2)
```

n= 686, number of events= 299

	coef	exp(coef)	se(coef)	z	Pr(> z)
age	-0.014507	0.985597	0.010065	-1.441	0.149478
pnodes	0.050122	1.051400	0.007422	6.753	1.45e-11 ***
horThyes	-0.711866	0.490728	0.754075	-0.944	0.345157
tgrade.L	0.702525	2.018843	0.185994	3.777	0.000159 ***
tgrade.Q	-0.213184	0.808007	0.121788	-1.750	0.080039 .
menostatPost	0.338578	1.402951	0.181505	1.865	0.062127 .
tsize	0.007605	1.007634	0.003931	1.935	0.053006 .
age:horThyes	0.006427	1.006448	0.013472	0.477	0.633322

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
exp(coef) exp(-coef) lower .95 upper .95
```

age	0.9856	1.0146	0.9663	1.005
pnodes	1.0514	0.9511	1.0362	1.067
horThyes	0.4907	2.0378	0.1119	2.151
tgrade.L	2.0188	0.4953	1.4021	2.907
tgrade.Q	0.8080	1.2376	0.6364	1.026
menostatPost	1.4030	0.7128	0.9830	2.002
tsize	1.0076	0.9924	0.9999	1.015
age:horThyes	1.0064	0.9936	0.9802	1.033

Concordance= 0.676 (se = 0.015)

Likelihood ratio test= 83.67 on 8 df, p=9e-15

Wald test = 101.3 on 8 df, p=<2e-16

Score (logrank) test = 107.4 on 8 df, p=<2e-16

1. Without using the results, explain what type of flexibility this interaction adds to the model.

Adding this interaction term gives our Cox model a lot of flexibility because without it the model would struggle to cope with factors that increase the risk ratio but do so inconsistently across groups. This one specifically enables the model to cope with the fact that hormone therapy might be slightly more helpful for older people.

2. Does the interaction appear important? How do you assess that?

This interaction doesn't appear important, which we can assess using the fact that the confidence interval for the exponentiated coefficient straddles zero and has a p value that fails to reject the null.

Part 3 — Model Diagnostics

7. Proportional Hazards Assumption

Test the proportional hazards assumption:

```
zph3 <- cox.zph(fit3)
zph3
```

	chisq	df	p
age	8.828	1	0.0030
pnodes	0.535	1	0.4646
horTh	0.166	1	0.6835
tgrade	11.554	2	0.0031
menostat	5.000	1	0.0254
tsize	0.257	1	0.6122
age:horTh	0.383	1	0.5362
GLOBAL	21.522	8	0.0059

1. Report results of `cox.zph()`

The test appears to indicate that the null hypothesis that the proportional hazards assumption is met can be rejected for age, tgrade, and menostat.

2. Comment on whether the PH assumption appears reasonable

I don't think that the proportional hazards assumption is completely reasonable in this situation because I can think of a lot of times when one of our factors would increase hazards non-proportionally across ages or grades of breast cancer. For instance, the same size or count of tumors might be more or less hazardous depending on how old someone is and what grade the tumors are considered to be.

Part 4 — Final Model and Interpretation

8. Final Model Selection

Choose a final model.

Justify your choice using:

- Model comparison
- Interpretability
- Diagnostics

Please discuss how you chose to strike a balance between model simplicity, assumptions, and comparison.

I'm going to go with model 1 because it doesn't do that much worse than model 2, and leaves out two of the variables that violate the proportional hazards assumptions. Plus it's a lot simpler and easy to interpret without any interactions.

9. Summary

Write a short summary (3–5 sentences):

- What variables are most important?
- What did you learn about survival in this dataset?
- Any limitations or concerns?

The most important variables consistently seem to be whether a person has gotten hormone therapy, the number of positive nodes, and the grade of those tumors. I learned that survival isn't always straightforward but it is possible to establish relationships between variables and proportional risk factors over time. The only concern that I still have is how to effectively handle the age variable since it seems important to but might only be important as far as it interacts with other variables. Further investigation into these interactions could be profitable.